Lab Assignment - 7

Computer Architecture (CS F342)
Semester-I, 2026-27
Department of Computer Science and Information Systems (CS & IS)
BITS-Pilani, K K Birla Goa Campus, Goa, India.

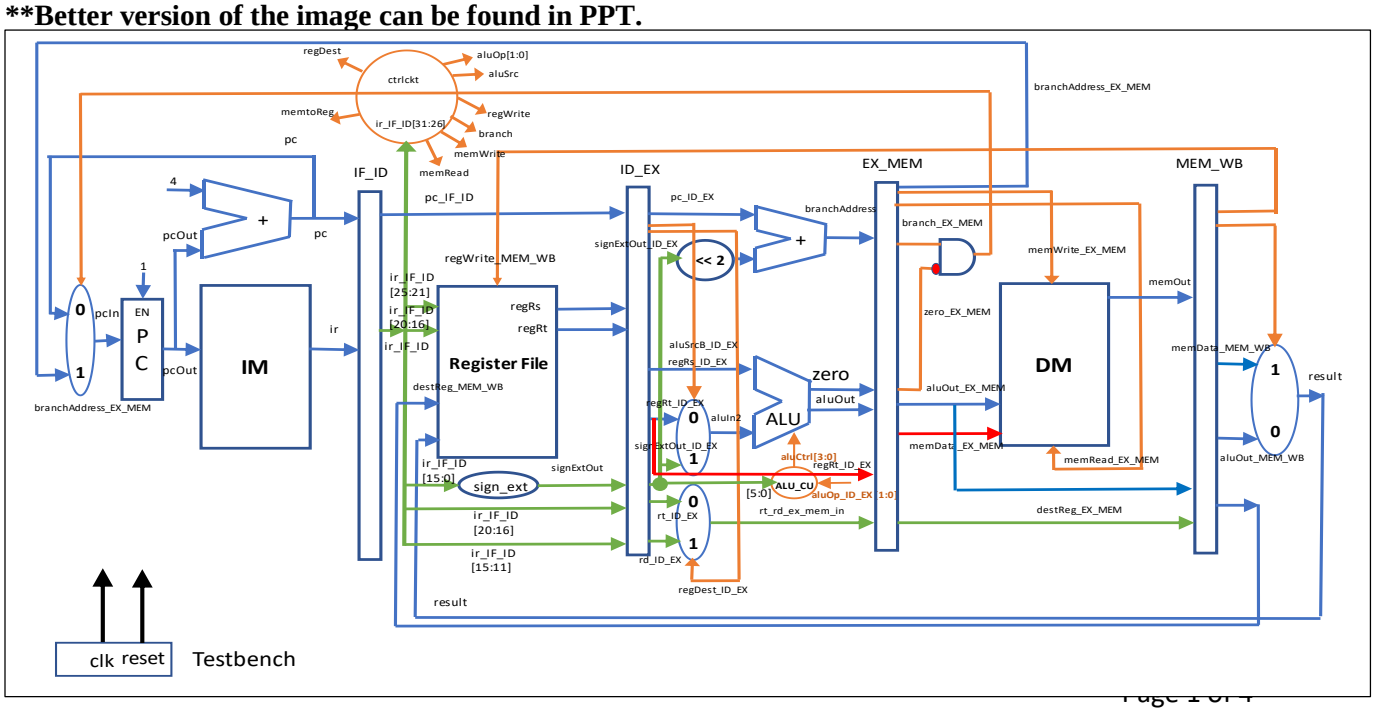
Due date: Oct 15, 2026: 13:00 Marks: 20

Consider the 32-bit MIPS instruction model (instruction format is given below) and, using Verilog HDL language, simulate the MIPS's instructions: LW, SW, ADD, ADDI, and BNE. For this, one has to define the following modules. There is no need to define other variables and modify the other modules except the below in the program.

File Name	Module	Marks
Dff.v	Dff_Reg	1
1bitReg.v	register1bit	1
2bitsReg.v	register2bit	1
5bitsReg.v	register5bit	1
stageIF_ID_Regs.v	IF_ID	1
stageID_EX_Regs.v	ID_EX	2
stageEX_MEM_Regs.v	EX_MEM	1
stageMEM_WB_Regs.v	MEM_WB	1
controller.v	controlCircuit	3
topModule.v	pipelinedDatapath	7
testbench.v	testbench	1

MIPS's pipelined model

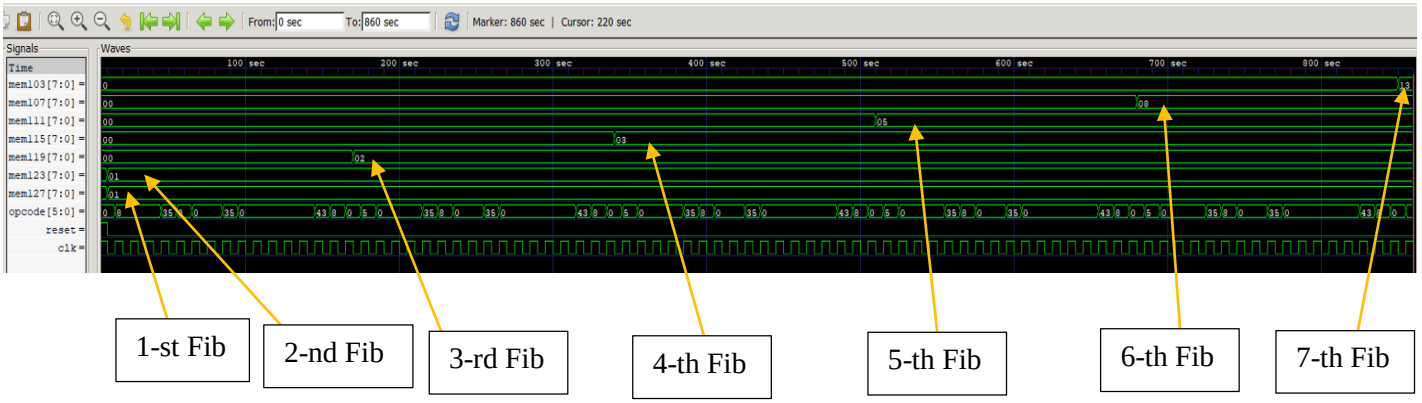
****Better version of the image can be found in PPT.**



Instruction format:

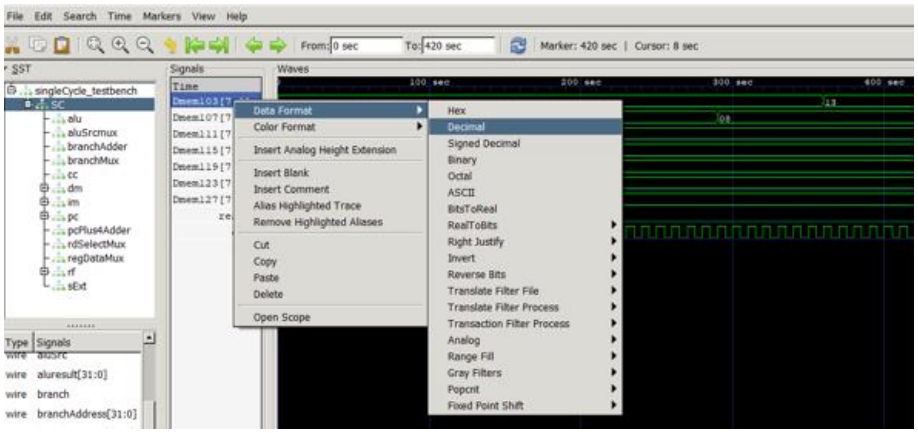
R-Type	**OPCODE**						**RS**					**RT**					**RD**					**SHAMT**					**FUNCT**				
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
I-Type	**OPCODE**						**RS**					**RT/RD**					**IMMEDIATE**															
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

In the template, a test program, stored in the memory, generates 7 Fibonacci numbers. The program is written using the above-mentioned instructions. NOP instruction is used to resolve data and control hazards. NOP is an instruction that doesn't affect the processor's states. The MIPS has no dedicated NOP instruction in the ISA. Here, *ADD R0, R0, R0* has been used as NOP instruction. The first two Fibonacci numbers are stored in the memory at locations 124 and 120. The program calculates the first seven Fibonacci numbers, and the remaining five Fibonacci numbers can be stored in locations 116 onwards. The Big-endian format is considered to store the instruction and data. Memory is byte-addressable. The memory contains 128 locations. However, the program counter (PC) is 32-bit. Some bits of the PC are used to identify the memory locations. Similarly, for ALU's output. The program generates the following outputs:



****Better version of the image can be found in PPT.**

Following is changing procedure for signals' data format as decimal:



Instructions for uploading the results:

Download the *CA_Lab-7_Template.zip* and rename the folder as *<CampusID>_CA_ Lab-7*. Output must be stored as *<CampusID>_ PipelineMIPS_output.png* in the corresponding folder.

Create a zip file *<CampusID>_CA_ Lab-7.zip*, which contains all the files mentioned above (modified), and submit it in Quanta. Do not create archives in other formats (rar, tar.gz etc). Once uploaded on Quanta, remember to submit for grading. Do not leave it as a draft.

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Note: Please don't upload the assignments, template file/solution and lab. manual on GitHub or others public repository.

Kindly remove them, if you have uploaded the previous assignments.

It violates the BITS's Intellectual Property Rights (IPR).

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